DOCTORAL PROGRAM IN DEEP TECH WITH A FOCUS ON ARTIFICIAL INTELLIGENCE AND EMERGING TECHNOLOGIES

Session 1 deliverable: Paradigm Justification Statement

Course name : Research Methods and Scientific Integrity in AI and Advanced Technologies
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1. Research topic and context, 3–4 sentences;

Peruvian public entities use information systems to support their mission-critical and support processes, and they increasingly incorporate artificial intelligence (AI) into their development, even though this integration remains largely unstructured and weakly governed. This research studies a reference architecture based on specialized generative AI agents to support the development and verification of information systems in regulated contexts. The proposal incorporates AI as a governed architectural component, establishes a conformance principle to verify compliance with constraints and regulations, and separates the generation and verification functions through independent agents. The model will be evaluated in the context of the Peruvian pension system, considering its contribution to the quality and timeliness of services, the protection of personal data, and the continuity of essential public services. Because pension-system data is highly sensitive, the conformance framework is anchored in the applicable regulations -Law No. 29733 on the Protection of Personal Data, Law No. 31814 promoting the use of artificial intelligence, and the NTP-ISO/IEC 42001:2025 standard on AI management systems- and it is explicitly established that the artifact operates in the development and verification phases, and not in production execution; moreover, the agents will run preferably in local, controlled environments, with access restricted to only the strictly necessary code and data, in order to preserve confidentiality and the continuity of services.

2. Preliminary research question, 1–2 sentences;

How should a reference architecture based on specialized generative AI agents be structured to automate and verify the development of critical state information systems, ensuring their compliance with the enterprise architecture and applicable regulations?

3. Chosen paradigm and justification, 5 to 7 sentences, explicitly ruling out the other three alternatives with specific reasons;

This research adopts the Design Science Research (DSR) approach because it seeks to design and evaluate an artifact to solve a specific software-engineering problem in regulated contexts. DSR is assumed here as a research approach -and not as an ontology- grounded in a pragmatist perspective that privileges the usefulness and applicability of the knowledge generated to solve concrete problems through the design and evaluation of artifacts (Hevner et al., 2004; Peffers et al., 2007). The positivist approach is ruled out because the main objective is not to measure an existing phenomenon, but to design and evaluate a solution. The interpretivist approach is also ruled out, since the research does not primarily seek to understand the perceptions or meanings of the actors involved. Moreover, it goes beyond simply describing the current adoption of AI in public entities. Mixed methods, in turn, do not constitute a competing paradigm, but rather an evaluation strategy available within DSR itself, combining qualitative and quantitative evidence to assess the artifact (Gregor & Hevner, 2013). The contribution consists of a reference architecture that integrates generative AI agents and compliance-verification mechanisms, evaluated through an application case in the Peruvian State's social security system.

4. Implications of this choice, 3–4 sentences in the form of a checklist;

This choice implies that: (i) a reference-architecture artifact will be designed based on enterprise-architecture principles, incorporating AI agents as governed components; (ii) a conformance-

verification mechanism will be defined to automatically assess compliance with architectural and regulatory constraints, preferably through deterministic techniques (static analysis, policy as code, and rule-checking against the enterprise architecture) so that verification does not depend solely on one generative model detecting another's errors; (iii) metrics will be established to evaluate the performance of the generation and verification process—specifically, the detection of conformance violations (precision and recall), the defect escape rate, rework effort, and lead time—assessed against a baseline of single-agent generation without independent verification, which isolates the contribution of the proposed separation; (iv) the artifact will be evaluated through an application case in a public entity, seeking to determine its usefulness for improving the quality and efficiency of digital-service development, complemented by a scenario-based architecture evaluation with expert participation and, if feasible, a second application case, so that the generalization from instance to reference architecture is evidenced rather than merely asserted; (v) the data sources that will underpin the design and evaluation will be specified (enterprise-architecture documentation, code repositories, and development artifacts from the case), restricting the use of personal data from the pension system; and (vi) the results are expected to be disseminated in relevant venues, including IEEE Software, Journal of Systems and Software, and Government Information Quarterly given the public-sector angle, as well as DESRIST or the European Conference on Software Architecture (ECSA) for the design contribution.

5. One doubt or tension, 1–2 sentences.

The main uncertainty lies in continued access to, and participation with, the entity that manages the Peruvian pension system during the design and evaluation phases of the research. If these conditions change, it will be necessary to identify an alternative organization that allows the artifact to be applied and evaluated under comparable conditions. A second tension is the independence between the generation and verification agents: if both are generative models, they could share training data, biases, and blind spots, so that their failure modes would be correlated and not fully independent; to mitigate this, the conformance layer will be made at least partially deterministic, and empirically demonstrating such independence would itself constitute a publishable result.

References

Gregor, S., & Hevner, A. R. (2013). Positioning and Presenting Design Science Research for Maximum Impact. MIS Quarterly, 37(2), 337–355.

Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design Science in Information Systems Research. MIS Quarterly, 28(1), 75–105.

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A Design Science Research Methodology for Information Systems Research. Journal of Management Information Systems, 24(3), 45–77.